Finite State Machine (FSM) Assignment

Subject: Digital Logic Design

Q1. What is a Finite State Machine (FSM)? Explain the difference between sequential circuits and combinational circuits. Also explain the basic components of an FSM.

Hints:
- State
- Input
- Output
- Clock
- Memory element

Q2. Explain Moore Machine and Mealy Machine with diagrams. On what factors does the output depend in both machines?

Also explain:
- Advantages
- Disadvantages
- Real-life applications

Q3. Write a detailed comparison between Moore Machine and Mealy Machine.

Comparison points:
- Output dependency
- Number of states
- Speed
- Stability
- Hardware complexity

Q4. Design the state diagram and state table of an FSM that detects the binary sequence “101”.

• State diagram
• State table
• Output table

Q5. Design a Moore Machine sequence detector for the binary sequence “110”.

• State diagram
• State table
• Output equations
• Explain why it is a Moore machine

Q6. Design a Mealy Machine sequence detector for the binary sequence “1011”.

• State transitions
• State diagram
• State table
• Output conditions
• Explain why it is a Mealy machine

Q7. Explain the procedure for converting a Mealy Machine into a Moore Machine and a Moore Machine into a Mealy Machine with suitable examples.

Include:
• Conversion rules
• State splitting
• Advantages of conversion
• Example state diagrams